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Slot - C

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Computer vision.

①

a) The computer vision enables computers to capture, process, and analyze images or videos so the can identify the objects.

⑥ The main objects of computer vision are to recognize objects, understand visual scenes

⇒ Two real-world application in computer vision are face recognition system.

②

⑥ Feature extraction is the process of identifying important characteristics of an image.

⑥ object recognition improves computer vision system by enabling them to identify and classify objects by accurately.

③

⑥ low level image processing involves basic operators such as image enhancement, noise

⇒ noise removal

⇒ filtering

⇒ contrast adjustment to improve image quality etc...

⑥ mid-level image processing focuses on segmenting image and extracting features, whereas high-level image processing interprets.

(4)

- (a) The computer vision pixel is the smallest unit of a digital image that represents a simple point with a specific intensity or color value.
- (b) Image resolution determines the number of pixels in an image. High resolution provides more detail and better quality.

(5)

- (a) An image sensor captures light from the scene and converts it into electrical signals that form the digital image.
- (b) Two image acquisition devices are a digital camera and a scanner.

(6)

- (a) Sampling is the process of converting a continuous image into a discrete grid of pixels.
- (b) Quantization assigns a finite number of intensities or color values to each sampled pixel.

(7)

- (a) Perspective Projection is the process of projecting a three-dimensional scene.
- (b) Illumination provides the light required to make objects visible and directly.

8

(a) Image Processing is the technique of enhancing, restoring, or modifying digital images to improve their quality.

(b)

Image processing improve image quality, Computer vision interpret image content.

9

(a)

Computer vision analyzes medical images such as x-rays, CT scans, and MRIs to detect diseases.

(b)

It detects roads, traffic, signs, pedestrians, and obstacles.

10

(a) ~~the~~ every digital image is composed of pixels, each containing brightness or color information.

(b)

Sampling determines the number of pixels, while resolution determines image detail.